



Term Evaluation (Even) Semester Examination March 2025

Roll no.....

Name of the Course and semester: **B. Tech CSE IV Core, Int., AI/DS, CS | 4th**

Name of the Paper: **Finite Automata and Formal Languages**

Paper Code: **TCS-402**

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub questions
- (ii) Each question carries 10 marks.
- (iii) Please specify COs against each question.

Q1. (10 Marks) [CO-2]

- a. Design a DFA over the input alphabet $\Sigma = \{a, b\}$ such that it does not accept the strings ending with either "aab" or "aba".

OR

- b. Design a DFA over the input alphabet $\Sigma = \{0, 1\}$ such that it accepts only the binary strings whose decimal equivalent is divisible by 5.

[For Example, 101, 0101, 1010, 1111, etc. are divisible by 5 so all these inputs should be accepted while, 100, 1100, 0111, etc. are not divisible by 5 so should be rejected]

Q2. (10 Marks) [CO-1]

- a. Apply Myhill-Nerode theorem to minimize the given DFA:

Q / Σ	0	1
$\rightarrow q_0$	q_1	q_2
q_1	q_3	q_5
q_2	q_5	q_4
$*q_3$	q_3	q_3
$*q_4$	q_4	q_4
q_5	q_5	q_5

OR

- b. Convert the given ϵ -NFA into an equivalent DFA where $\Sigma = \{0, 1, 2\}$:

State/ symbol	ϵ	0	1	2
$\rightarrow p$	$\{q, r\}$	-	$\{q\}$	$\{r\}$
q	-	$\{p\}$	$\{r\}$	$\{p, q\}$
$*r$	-	-	-	-

Q3. (10 Marks) [CO-2]

- a. Design a Moore machine as a sequence detector over the input alphabet $\Sigma = \{a, b\}$. The Moore machine should generate a "1" whenever there is "abb" in the input sequence otherwise nothing.

[Samples are: Input: abbaabbaa, Output: 11

Input: abbbabbbabb, Output: 111]

OR



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- b. Convert the given Mealy machine to an equivalent Moore machine. Also, Test the output of Mealy and Moore machine for the input "101101" to prove the equivalence.

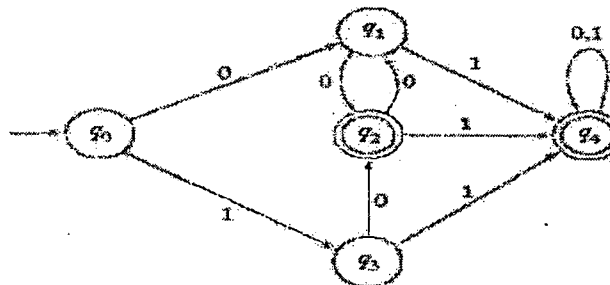
Present State	Input = 0		Input = 1	
	Next State	Output	Next State	Output
→q ₁	q ₃	A	q ₂	A
q ₂	q ₁	A	q ₄	B
q ₃	q ₂	B	q ₁	A
q ₄	q ₄	A	q ₃	B

Q4. (10 Marks) [CO-2]

- a. Define ϵ -closure (Epsilon-closure). State and prove Kleene's Theorem for showing the equivalence between Regular Expression and Finite Automata.

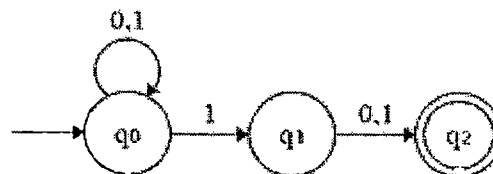
OR

- b. Find the regular expression for the given finite automata using Arden's theorem:



Q5. (10 Marks) [CO-1]

- a. Define NFA (Non-Deterministic Finite Automata). Convert the given NFA to equivalent DFA:



OR

- b. State pumping lemma for regular languages. Prove that the given language L is non-regular:
 $L = \{a^i b^j c^k \mid k > i+j\}$